

VINCENT DO

St. Louis, MO · 779-207-6818 · vincentdo306@gmail.com · [linkedin.com/in/vincent-do-uiuc](https://www.linkedin.com/in/vincent-do-uiuc) · github.com/vincentdo1 · vmd306.com

EXPERIENCE

Boeing Company

Aug 2024 – Present

Software Engineer

Berkeley, MO

- **Designed** the message protocol, internal SDK, and service interface for a real-time messaging framework—**adopted by other engineers** as the standard contract for downstream modules
- Shipped production C++ features delivering **6+ real-time navigation message streams** to embedded display systems, with automated tests and Dockerized build workflows powering end-to-end deployment
- Own integration and release readiness for a shared framework consumed by **60+ dependent modules**, eliminating compatibility regressions by resolving binary timing drifts under real-time constraints
- Resolved a long-standing cross-system defect **within 2 weeks of onboarding** to unfamiliar systems by tracing a binary data merge edge case, restoring correct downstream output behavior
- Led month-long evaluation of partner software through custom integration with our messaging architecture, validating customer requirements at **1.5× the required throughput** and documenting procedures for future releases

Expedia Group

May 2023 – Aug 2023

Software Development Engineer Intern

Chicago, IL

- Shipped customer-facing Expedia mobile search features—**VIP Access badges (A/B-tested with measurable lift on bookings and CTR)**, **20+ amenity filters**, and error-handling improvements—reducing booking search friction
- Implemented GraphQL schema changes across **5+ Kotlin microservices**, enabling new mobile booking flows across backend and client surfaces
- Built backend support for localized search components across **16 languages**, partnering with frontend and localization teams to validate dynamic rendering before release

University of Wisconsin–Madison

May 2021 – Aug 2021

Biomedical Undergraduate Researcher

Madison, WI

- Analyzed 16,813 genes across 169 phenotypes using Python and R with graph traversal and visualization to identify phenotype-gene associations
- Narrowed the candidate space to 25 high-priority Zellweger genes via frequency thresholds, chromosomal-location filtering, and recursive phenotype hierarchy traversal

PROJECTS

Volleyball Motion Tracker | *Python, MediaPipe, YOLOv8, OpenCV, NumPy*

[volleyball-motion-tracker](#)

- Built a per-frame volleyball spike analyzer fusing pose tracking and ball detection to extract **13+ kinematic and force metrics per spike**, shipped as a pip-installable, unit-tested CLI
- Derived monocular biomechanics from a **single uncalibrated video**—anthropometric scale calibration, jump height from hip trajectory, and ball-impact force from a contact-time × velocity-delta model

Playable Chess AI | *Python, PyTorch, CUDA, Flask, JavaScript*

[live demo](#) | [code](#)

- Trained a ResNet + LSTM policy-value model on 4.18M GM/Magnus PGN positions, reaching 71.2% top-5 accuracy on a 390K-position held-out test set, with a legal-move-policy head and temperature sampling for variety
- Engineered a GPU-accelerated PyTorch training pipeline using mixed precision and a custom IterableDataset, cutting per-epoch time from 20+ hours on CPU to 15–25 minutes
- Deployed an ES-modular browser frontend on GitHub Pages plus a Flask inference API on Railway, supporting Human/Stockfish-WASM/Random/Alphabeta/NN players with sub-second response times

Shortest Path Between Airports | *C++, React, WebGL*

[live demo](#) | [code](#)

- Implemented BFS, Floyd–Warshall, and betweenness centrality over the OpenFlights aviation graph (**1,000+ airports**) to rank the most influential nodes, validated by C++ unit tests across all graph operations
- Designed a WebGL globe frontend with a refactored feature-based module structure and a GitHub Actions workflow **auto-deploying every push** to GitHub Pages

TECHNICAL SKILLS

Languages: C++, Python, Java, C#, JavaScript, TypeScript, Kotlin, SQL

Frameworks/Libraries: React, Node.js, Flask, PyTorch, GraphQL, RESTful APIs

Tools/Platforms: Docker, Jenkins, Git, Linux, GCP, PostgreSQL, MongoDB, Railway, GitHub Actions, CI/CD

EDUCATION

University of Illinois Urbana-Champaign

Aug 2020 – May 2024

B.S. Computer Science & Chemistry · GPA: 3.6

Champaign, IL